PARISHRAM 2.0

CLASS 12TH 2026

Relations and Functions

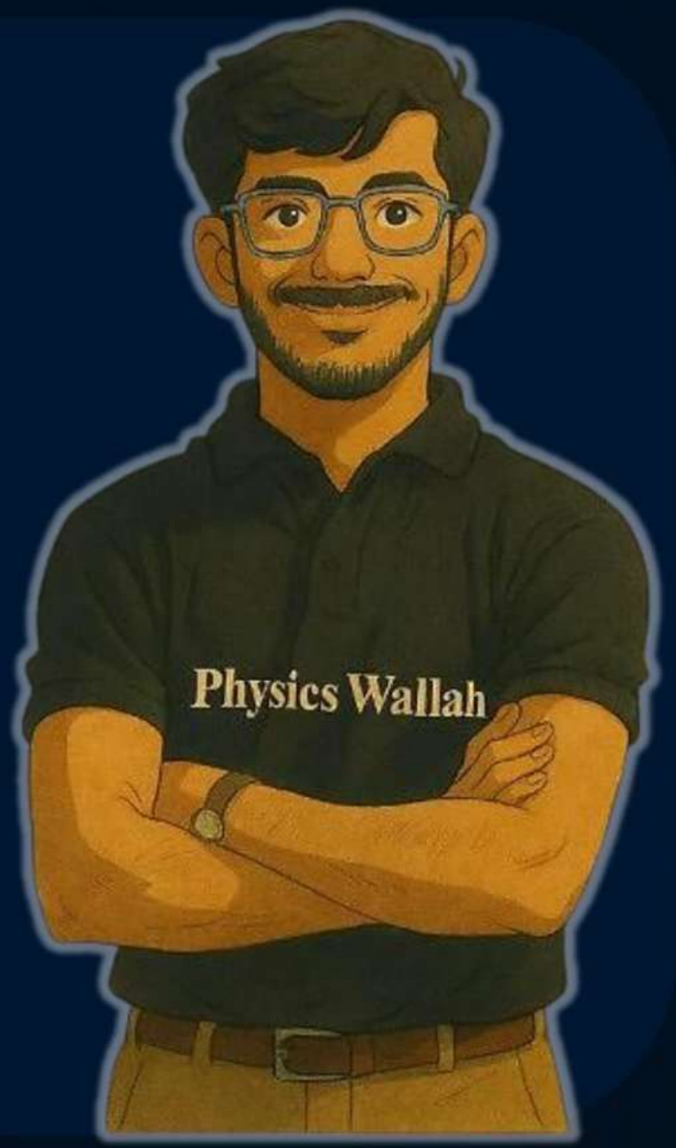
MATHS **LECTURE-2**

BY-MANDAR SIR



TOPICS TO BE COVERED

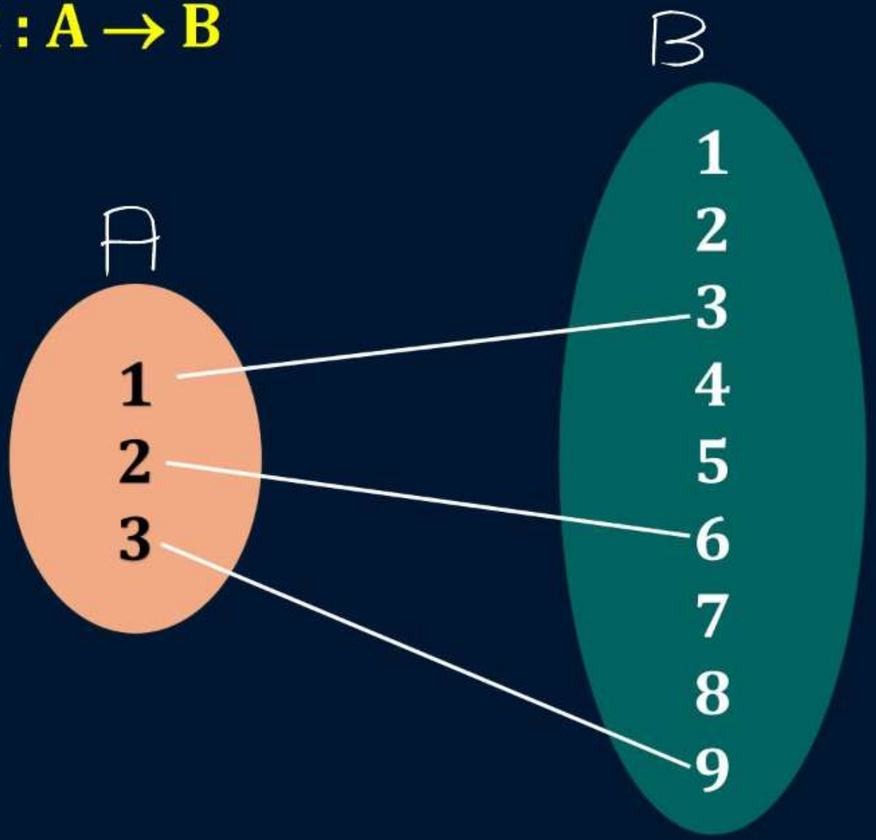
1. Inverse Relations
2. Types of Relations
3. Questions Based on it



Recap

DOMAIN, RANGE, CO-DOMAIN

R : A → B



Domain → input

Range → output

Co-domain → ↖

R : A → B

B (Saare elements)

R : B → A

$R = \{(1, 3)(2, 6)(3, 9)\}$

Domain = {1, 2, 3}

Range = {3, 6, 9}

Co-domain → B ke saare elements

↳ {1, 2, 3, 4, 5, 6, 7, 8, 9}

POLL QUESTION

If A is the set of even natural numbers less than 8 and B is the set of prime numbers less than 7, then the number of relation form A to B is

$A = \{2, 4, 6\}$ $\quad n(A) = 3$

$B = \{2, 3, 5\}$ $\quad n(B) = 3$

No of Relations $(A \to B) = 2^{n(A) \times n(B)}$

$= 2^{3 \times 3}$

$= 2^9$

1. 2^9 ✓
2. 9^2
3. 3^2
4. $2^9 - 1$

Inverse Relation

Let R be a relation from A to B, then R^{-1} from B to A is defined as the inverse of R

i.e. $R = \{(x, y)$

$R^{-1} = \{(y, x) : (x, y) \in R\} \subseteq B \times A$

$= \quad R = \{(2, 3), (4, 5), (-6, -2)\}$

$\Rightarrow \; R^{-1} = \{(3, 2), (5, 4), (-2, -6)\}$

$R = \{(1, 2)(2, 3), (4, 5)\}$

$R^{-1} = \{(2, 1), (3, 2), (5, 4)\}$

Empty Relation

A Relation R in a set A is called empty relation, if no element of A is related to any element of A,

i.e., $R = \phi \subset A \times A$

$R = \phi$ $\quad$ $\phi \rightarrow$ Null

$(1,1)(1,2)(1,3)(2,1)(2,2)$

Example : Consider the relation R on the set $A = \{1, 2, 3\}$ given by

$R = \{(a, b) : a, b \in A, a + b = 10\}$

$R = \phi$

Identity Relation → (Sirf Twins hone chiye)

A relation R is said to be identity relation if

$$R = \{(x, x) : x \in A\}$$

$$R = \{(x, y) : x = y, x \in A \text{ and } y \in B\}$$

$A = \{1, 2, 3\}$

$R_1 = \{(1,1)(2,2)(3,3)\}$ ✓

$R_2 = \{(1,1),(2,2),(3,3),(1,2)\}$ ✗

↑ Twin ke alawa

$B = \{a, b\}$

$R = \{(a,a),(b,b)\}$ ✓

Universal Relation

A relation R in a given non-empty set i.e. $R \subseteq A \times A$ is called universal relation if $R = A \times A$

$A = \{1, 2\}$ $A = \{1, 2\}$

$A \times A = \{(1,1)(1,2)(2,1)(2,2)\}$

$R = \{(1,1)(1,2)(2,1),(2,2)\}$

$R = A \times A$

Agar R mea $A \times A$ (saare elements) } Universal Relation

Reflexive Relation

→ Twins
sab elements ke chiye

A relation in R is called reflexive if each element of set is related to itself.

$$R = \{(x, x) : x \in A\}$$

$A = \{1, 2\}$

$R_1 = \{(1,1)(2,2)\}$ Reflexive ✓ Identity ✓

$R_2 = \{(1,1), (1,2), (2,2)\}$ Reflexive ✓ Identity ✗

$R = \{(1,1)(2,2)(3,3)\}$

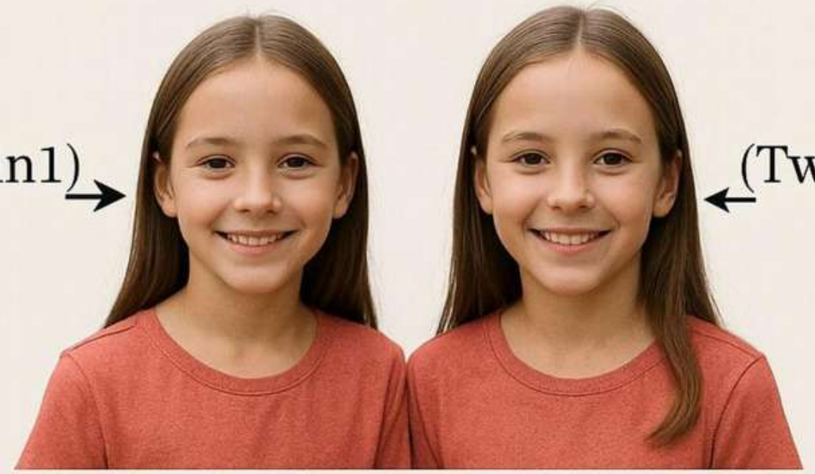
$A = \{1, 2, 3\}$

$R = \{(1,1)(2,2),(1,3)\}$

$(3,3) \rightarrow ??$

Reflexive
- → Yes
- → No

IDENTITY RELATION I_A



Only Twins (Saare elements)

REFLEXIVE RELATION R on A



Twins (Saare elements)

POLL QUESTION

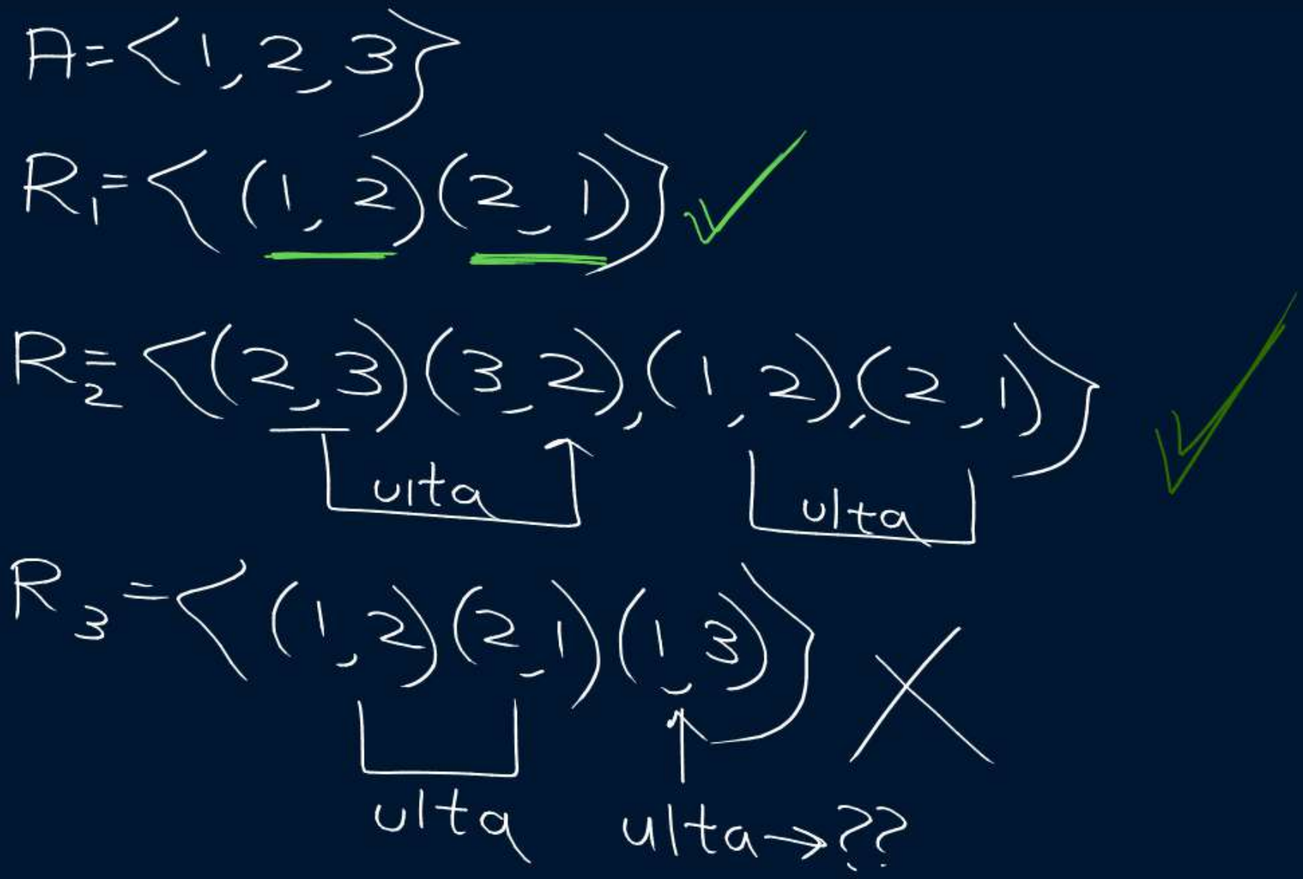
If a relation R is defined on let $A = \{1, 2, 3, 4\}$ relation. Which of the following is reflexive.

Twins chiye

1. $R = \{(1, 1), (2, 2), (1, 2), (2, 3), (3, 3), (4, 4)\}$ → Reflexive ✓
2. $R = \{(1, 1), (2, 2), (1, 2), (2, 4), (1, 4)\}$ ✗ → Identity (Sirf Twins) ✗
3. $R = \{(1, 1), (2, 3), (4, 4)\}$ ✗
4. $R = \{(1, 4), (4, 1)\}$ ✗

Symmetrical Relation

A relation R is said to be symmetric if $(x, y) \in R$ then $(y, x) \in R$.



POLL QUESTION

If a relation R is defined on let $A = \{a, b, c, d\}$. Which of the following is symmetric.

1. $R = \{(a,a), (a,b), (a,b), (b,c), (c,d)\}$

2. $R = \{(a,b), (b,b), (b,a), (a,c)\}$

3. $R = \{(a,a), (b,c), (c,b), (c,d), (d,c)\}$

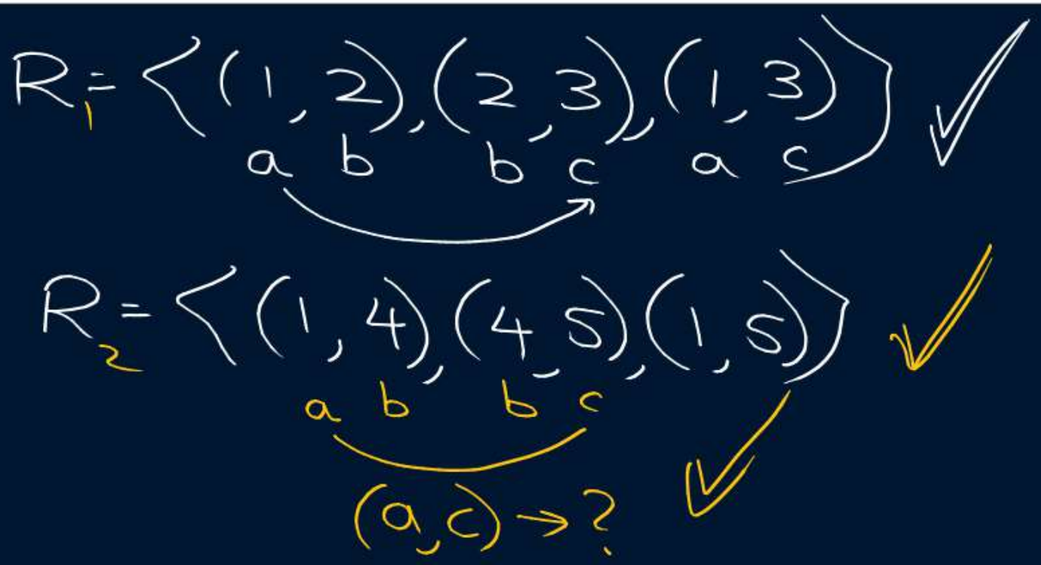
4. $R = \{(a,b), (b,c), (a,d)\}$

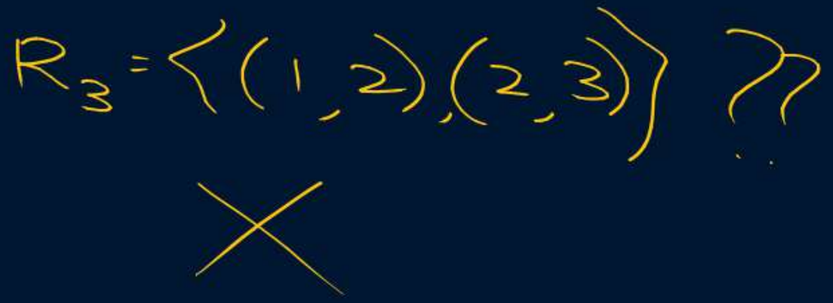
Transitive Relation

A relation R on a given non-empty set A is said to be transitive if

$(a, b) \in R$ & $(b, c) \in R$ then $(a, c) \in R$, where $a, b, c \in A$.

$R = \{(1, 2), (2, 3), (1, 3)\}$





QUESTION

Determine whether each of the following relations are reflexive, symmetric and transitive:

(i) Relation R in the set A ={1, 2, 3 ,.....,13, 14} defined as

$R = \{(x, y) : 3x - y = 0\}$

$R = \{(x, y) : 3x = y\}$

$R = \{(1, 3), (2, 6), (3, 9), (4, 12)\}$

(a, b) (b, c)

Reflexive → ✗ Sab ke Twins ✗

Symmetric → $(1, 3) \in R$, $(3, 1) \notin R$ ✗

Transitive → $(1, 3) \in R$ (a, b)

$(3, 9) \in R$ (b, c)

$(1, 9) \notin R$ ✗

QUESTION

Determine whether each of the following relations are reflexive, symmetric and transitive:

(ii) Relation R in the set N of natural numbers defined as

$R = \{(x, y) : y = x + 5 \text{ and } x < 4\}$

$R = \{(1,6)(2,7)(3,8)\}$

$a\ b \quad x\ y \quad v\ w$

Reflexive → Twin ✗ $(1,1) \notin R$ ✗

Symmetric → $(1,6) \in R$

$(6,1) \notin R$ ✗

Transitive → ✓

If

$(a,b) \in R$ ✓

$(b,c) \in R$ ✓

then

$(a,c) \in R$ ✓

QUESTION

Determine whether each of the following relations are reflexive, symmetric and transitive:

(iii) Relation R in the set $A = \{1, 2, 3, 4, 5, 6\}$ as

$R = \{(x, y) : y \text{ is divisible by } x\}$

QUESTION

Show that that relation R in the set R of real numbers, defined as $R = \{(a, b) : a \leq b^2\}$ is neither reflexive nor symmetric nor transitive.

Homework

Check For Reflexive, Symmetric & Transitive

(a) $R = \{(x, y) : x \text{ and } y \text{ work at the same place}\}$

(b) $R = \{(x, y) : x \text{ and } y \text{ live in the same locality}\}$

(c) $R = \{(x, y) : x \text{ is exactly 7 cm taller than } y\}$

(d) $R = \{(x, y) : x \text{ is wife of } y\}$

THANK YOU